

Optimizing Robotic Hand Morphology for Dexterous In-Hand Manipulation

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Abstract—We present an end-to-end optimization framework for humanoid robotic hand morphology, demonstrated on a complex in-hand cube reorientation task. The pipeline integrates parametric hand generation, reinforcement learning (RL) within a parallel physics simulator, and hyperparameter optimization. The search space encompasses the number of fingers, segment count, finger width scale, and fixed joint placement. Candidate morphologies are procedurally generated and evaluated in Isaac Lab using Proximal Policy Optimization (PPO). Proficiency is quantified by the Mean Episode Reward (R_{mean}), guiding the SMAC3 optimizer to propose subsequent configurations. Results confirm that optimized morphologies substantially improve task success, highlighting the critical interplay between physical design and control policy acquisition.

I. INTRODUCTION

Developing effective humanoid end-effectors for dexterous in-hand manipulation requires rigorous physical morphology investigation. While deep reinforcement learning (DRL) enables sophisticated control, policies are constrained by hardware capabilities. Since manual design is resource-intensive and often yields suboptimal solutions, we present an automated pipeline to generate, train, and evaluate robotic hand designs in simulation. By framing morphological design as a hyperparameter optimization (HPO) problem, the system autonomously explores structural configurations to maximize manipulation capabilities and task performance.

II. RELATED WORK

Our methodology intersects automated hardware design, DRL, and HPO.

DRL for Morphological Co-adaptation: DRL solves complex manipulation tasks [1], [2]. Recent studies show that human-like designs are not universally optimal, prompting interest in co-optimizing morphology and control [3]–[6]. However, joint optimization methods often face sample inefficiency and scalability issues in high-dimensional spaces. We address this by dictating physical structure based purely on task performance and evaluating configurations in Isaac Lab [7], a massively parallel simulator.

HPO in Hardware Design: HPO is an emerging hardware-software co-design strategy. Although traditionally used for machine learning models [8], Bayesian optimization and sequential model-based approaches effectively navigate complex search spaces. In robotics, multi-fidelity optimization techniques bypass the limitations of manual iteration by leveraging lower-fidelity evaluations to rapidly prune the search space.

III. IMPLEMENTATION

We use the Sequential Model-Based Algorithm Configuration (SMAC3) framework [9] to explore the structural search space (Fig. 1). SMAC3 utilizes random forest surrogate models and Bayesian optimization to propose configurations. To manage computational costs, we employ the SMAC3 MultiFidelityFacade to rapidly discard unpromising morphologies based on initial learning trajectories.

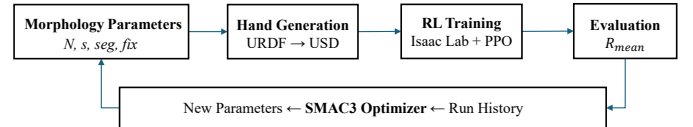


Fig. 1: Overview of the automated morphological optimization pipeline. SMAC3 optimizer continuously updates the structural hand parameters based on reinforcement learning performance evaluation, where N represents the number of fingers, s is the finger width scale, seg denotes the number of segments per finger, and fix indicates the location of fixed joint.

A. Morphological Search Space Definition

The discrete search space is defined by four variables:

- **Number of fingers (N):** Ranging from 2 to 6. Modified from an Allegro Hand, configurations with 2 to 5 fingers include one opposing thumb, while a 6-finger configuration features four regular fingers and two opposing thumbs. This determines maximum simultaneous contact points, directly impacting grasp stability and multi-fingered coordination.
- **Finger width scale (s):** Ranging from 0.5 to 1.5 in 0.1 increments. This alters frictional surface area, balancing increased grasp reliability against self-collision risks.
- **Segments per finger (seg):** Constrained to 3, 4, or 5 links. This dictates kinematic complexity, defining the hand's ability to wrap around object contours.
- **Fixed joint location (fix):** Designated at joint indices 0 through 4, or "none". Immobilizing a joint simplifies the control space but constrains theoretical dexterity. Invalid combinations (e.g., fixed joint index exceeding available joints) are explicitly excluded from the search space.

The palm width automatically adapts based on N and s .

B. Automated Generation and RL Training

A procedural generator constructs a URDF model based on proposed parameters, which is converted to a Universal Scene

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Description (USD) format for Isaac Sim. The generated model is deployed in an Isaac Lab cube reorientation environment. The agent is trained using PPO with a dense reward function prioritizing target orientation proximity.

C. Evaluation and SMAC3 Strategy

Manipulation capability is evaluated using the Mean Episode Reward (R_{mean}) over the training trajectory:

$$R_{\text{mean}} = \frac{1}{T} \sum_{t=1}^T r_t, \quad (1)$$

where r_t is the reward at step t and T is the total steps. This metric rewards rapid learning and high final proficiency. Using a Successive Halving bracket with a reduction factor (η) of 3, SMAC3 evaluates an initial batch of three configurations for 667 iterations each. The configuration with the highest R_{mean} is promoted to the full maximum budget of 2000 iterations. The complete run history updates the surrogate model to propose subsequent parameters.

IV. RESULTS

Extensive experiments across the parameter space provided comprehensive data on the impact of physical morphology.

A. General Performance

Mean reward curves spanning 2000 iterations were generated to isolate and analyze the impact of individual parameters. The data reveals distinct performance variations. Increasing the number of segments from 3 to 5 generally resulted in a higher mean reward trajectory, suggesting higher kinematic complexity is beneficial for this specific task.

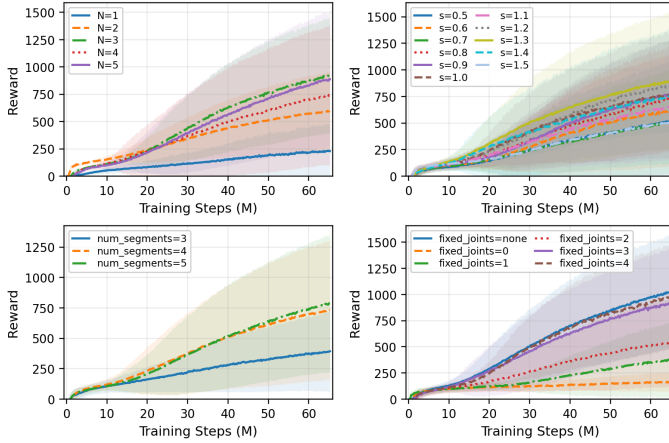


Fig. 2: Grouped reward curves by morphology parameters. Top-left: Number of fingers (N); top-right: Finger width scale (s); bottom-left: Segments per finger (seg); bottom-right: Fixed joint location (fix). Lines show grouped mean rewards; shaded regions indicate standard deviation.

Furthermore, our results demonstrate that configurations with 3, 4 or 5 fingers consistently outperformed those with fewer fingers, particularly when paired with larger finger scales (1.2 to 1.5).

B. Optimal Morphological Configurations

Some high-performing morphologies identified are:

- **Top** (R_{mean} **1385.88**): 4 fingers (3 regular, 1 thumb), scale 1.4, 5 segments, fixed joint between segments 2 and 3.
- **Second** (R_{mean} **1303.90**): 5 fingers (4 regular, 1 thumb), scale 1.3, 4 segments, fully actuated.
- **Notable Alternative** (R_{mean} **1143.44**): 3 fingers (2 regular, 1 thumb), scale 0.5, 5 segments, fully actuated.
- **Baseline: Allegro Hand** (R_{mean} **462.57**): 4 fingers (3 regular, 1 thumb), scale 1.0, 4 segments, fully actuated.

Conversely, morphological variations restricted to two fingers (one regular finger and one thumb) consistently failed to achieve meaningful manipulation, reflecting minimal mean reward values regardless of the selected scale or segment count.

V. DISCUSSION AND CONCLUSION

The pipeline successfully automates design optimization of robotic hands for a cube reorientation task. Results confirm that physical morphology significantly impacts reinforcement learning efficacy and overall manipulation success. Additionally, SMAC3's multi-fidelity approach enabled efficient exploration of a complex physical parameter space.

Future work expands the search space to include asymmetric finger placements and variable palm geometries. We will explore alternative optimization strategies, such as evolutionary methods, and assess generalizability across diverse in-hand manipulation tasks.

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